

Proficiency-Test-II

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Task-1

```
for(i in 1:3){  
  print(mtcars$mpg[i])  
}
```

```
[1] 21  
[1] 21  
[1] 22.8
```

Task-2

```
count_above <- function(vector, cutoff){  
  # create a function named count_above with two inputs: the numeric vector and the cutoff value.  
  count <- 0  
  # initialize and set the value of count to 0.  
  for(i in 1:length(vector)){  
    # check every elements in the given vector.  
    if(vector[i] > cutoff){  
      # check if the element is larger than the cutoff value.  
      count <- count + 1  
      # if it satisfied the condition, count this element into count variable.  
    }  
  }  
  return(count)  
  # return the value of the observations.  
}
```

```
count_above(iris$Sepal.Length, 6)
```

```
[1] 61
```

```
count_above(mtcars$disp, 149.9)
```

```
[1] 20
```

Task-3

Data masking allows you to refer to variables in the “current” data frame (usually supplied in the `.data` argument), without any other prefix.

Task-4

```
iris$Sepal.Length[1]
```

```
[1] 5.1
```

```
iris[1, 1]
```

```
[1] 5.1
```

It returns the first iris flower’s sepal length.